

Smart Building AI Agent – Final Project Report

Project Title

Smart Building AI Agent for Energy-Efficient HVAC Optimization using EnergyPlus and Large Language Models

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1. Project Overview

The Smart Building AI Agent is an AI-driven closed-loop building energy optimization system that automatically improves HVAC operation while maintaining occupant thermal comfort.

Unlike traditional simulations where parameters are manually modified, this system continuously analyzes EnergyPlus simulation outputs, reasons using an LLM, updates HVAC parameters, regenerates building models, and reruns the simulation until an optimized solution is achieved.

The objective is to reduce total building energy consumption while ensuring acceptable thermal comfort measured using PMV (Predicted Mean Vote) and PPD (Predicted Percentage of Dissatisfied).

Note: Independent hackathon prototype. Not affiliated with, sponsored by, or endorsed by Honeywell International Inc. Just for hackathon purpose the name has Honeywell dashboard.

2. Problem Statement

Commercial buildings consume a significant amount of electricity through HVAC systems.

Existing building simulations require engineers to manually:

- inspect EnergyPlus outputs
- interpret comfort metrics
- modify HVAC settings
- rerun simulations

This process is slow, repetitive, and unsuitable for autonomous optimization.

Our solution automates the complete optimization loop using an intelligent AI agent.

3. Objectives

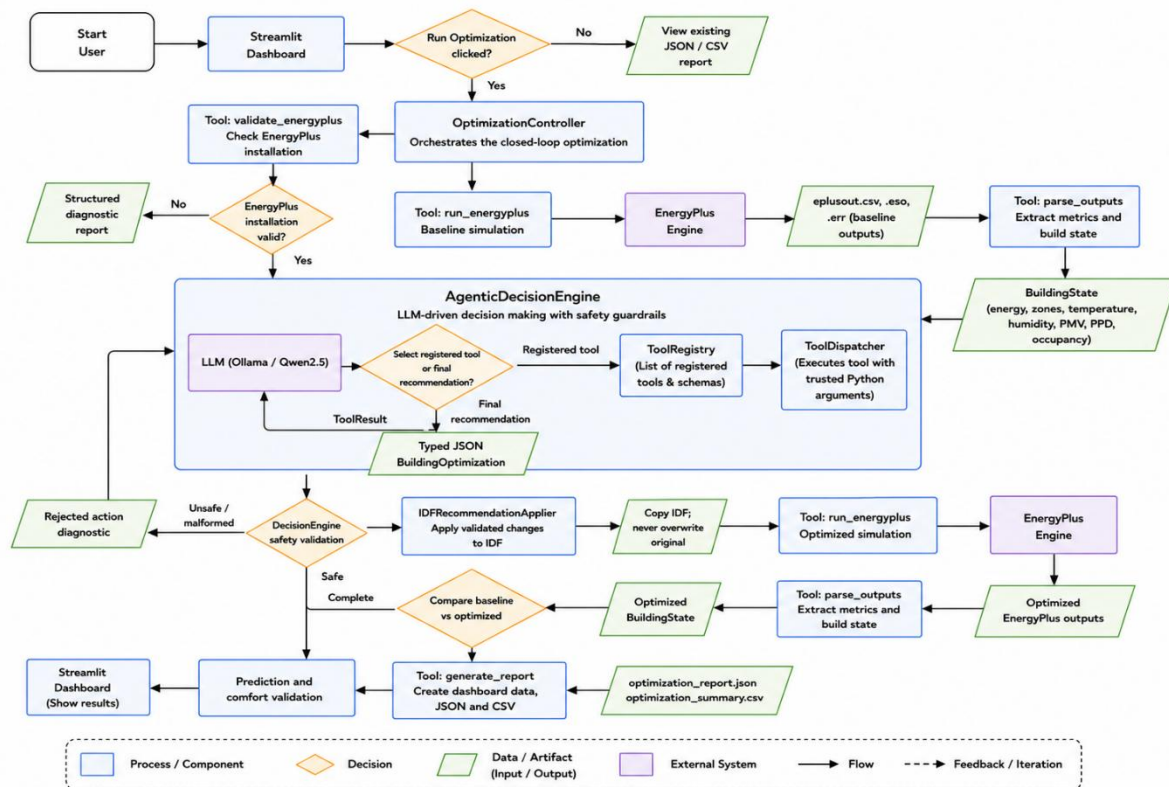
- Reduce HVAC energy consumption
- Maintain occupant comfort
- Automatically update EnergyPlus parameters
- Demonstrate AI-powered closed-loop optimization
- Provide visual comparison between baseline and optimized buildings

4. System Architecture

The project consists of the following components:

1. User Interface (Streamlit)
2. EnergyPlus Simulation Engine
3. Python EnergyPlus Wrapper
4. LLM Agent
5. Optimization Engine
6. IDF Modifier
7. Results Dashboard

Workflow:



5. Technologies Used

Technology	Purpose
Python	Core implementation
Streamlit	Dashboard
EnergyPlus	Building simulation
EnergyPlus API	Simulation control
Ollama qwen2.5 local AI reasoning	
Pandas	Data processing
Matplotlib	Visualization
Plotly	Interactive charts
JSON	Agent communication
Markdown	Documentation

6. AI Agent Workflow

The LLM agent performs the following tasks:

- Reads EnergyPlus outputs
- Evaluates HVAC performance
- Checks PMV values
- Checks PPD values
- Identifies energy-intensive components
- Generates optimization recommendations
- Updates HVAC control parameters
- Triggers the next simulation

The optimization continues until thermal comfort constraints are satisfied.

7. Thermal Comfort Metrics

PMV (Predicted Mean Vote)

PMV predicts how occupants perceive indoor temperature.

Range:

-3 = Cold

0 = Neutral

+3 = Hot

Target:

$-0.5 \leq \text{PMV} \leq +0.5$

PPD (Predicted Percentage of Dissatisfied)

PPD estimates the percentage of occupants likely to feel uncomfortable.

Target:

$\text{PPD} < 10\%$

8. Optimization Strategy

The AI agent automatically modifies:

- Cooling setpoint
- Heating setpoint
- Supply air temperature
- Fan schedules
- HVAC operating schedules
- Ventilation settings

Each modification generates a new IDF model which is simulated again.

9. Project Features

- ✓ Fully automated optimization
- ✓ Closed-loop AI control
- ✓ Automatic IDF generation
- ✓ Thermal comfort validation
- ✓ Energy savings dashboard
- ✓ Multiple optimization iterations
- ✓ Streamlit visualization
- ✓ Comparative analysis

10. Quantitative Results

The dashboard compares:

- Baseline Energy Consumption
- Optimized Energy Consumption
- Energy Savings (%)
- PMV
- PPD

The optimization demonstrates measurable reduction in total building energy consumption while maintaining thermal comfort within acceptable limits.

11. Repository Structure

smart-building-ai-agent/

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└─ app.py

└─ main.py

└─ requirements.txt

└─ README.md

└─ architecture.md

└─ optimization/

└─ energyplus/

└─ agent/

└─ dashboard/

└─ idf/

| └─ baseline.idf

| └─ optimized.idf

└─ results/

└─ outputs/

└─ assets/

12. Deliverables

1. Source Code

Complete Python implementation including:

- EnergyPlus wrapper
- AI agent
- Optimization engine
- Dashboard
- Communication pipeline

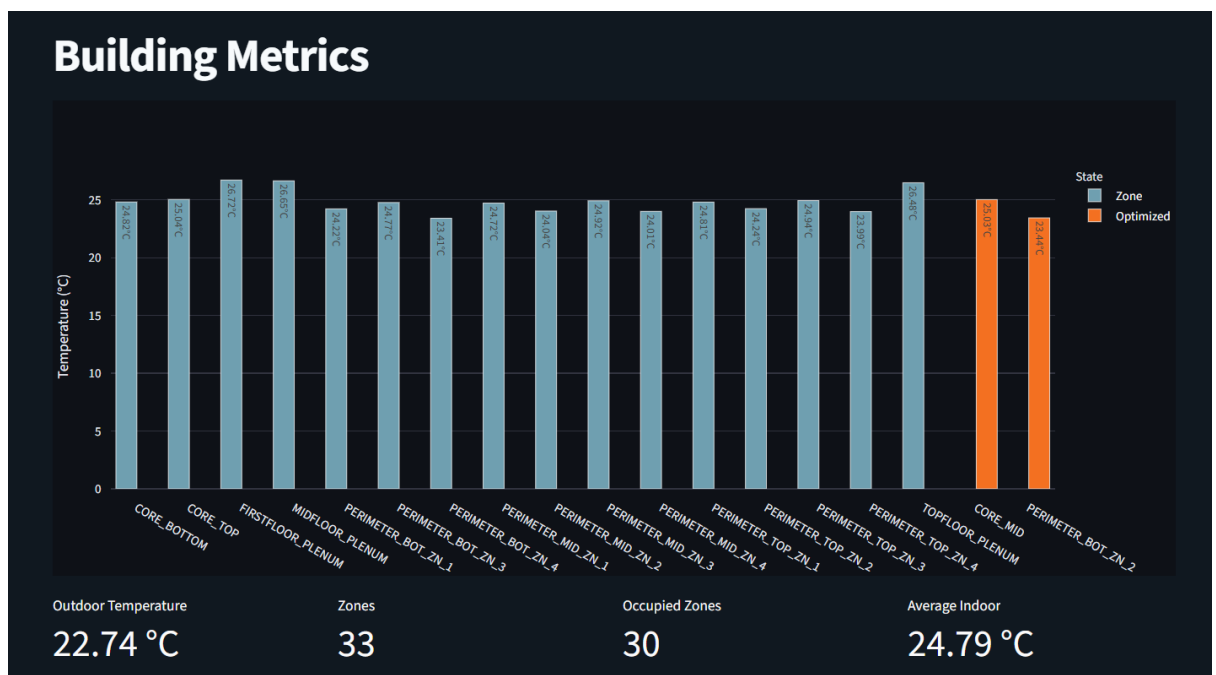
2. Building Models

- Baseline IDF
- Optimized IDF
- Runtime generated models

3. Dashboard

Displays:

- Energy consumption
- PMV
- PPD
- Percentage savings
- Baseline vs Optimized comparison



Zone details

Zone	Indoor Temperature (°C)
CORE_BOTTOM	24.8193
CORE_MID	25.0269
CORE_TOP	25.0411
FIRSTFLOOR_PLENUM	26.7172
MIDFLOOR_PLENUM	26.6496
PERIMETER_BOT_ZN_1	24.2248
PERIMETER_BOT_ZN_2	23.443
PERIMETER_BOT_ZN_3	24.7713
PERIMETER_BOT_ZN_4	23.4107
PERIMETER_MID_ZN_1	24.7233

HONEYWELL | BUILDING OPERATIONS CENTER

SMART BUILDING AI AGENT

Autonomous HVAC Energy Optimization using EnergyPlus + Agentic AI

EnergyPlus Connected

Ollama Connected

Agent Ready

Dashboard Overview

Autonomous building optimization control room

⚡ Baseline Energy 129.19B reported units	⚡ Optimized Energy 126.30B reported units	⬇️ Measured Energy Savings (EnergyPlus) 2.24% simulation result	⬆️ PMV -0.24 thermal comfort	● PPD 7.09% thermal comfort	🕒 Execution Time 33.58 s controller runtime
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MEASURED ENERGY SAVED (ENERGYPLUS)

2.24%

⬇️ 2,890,126,312.90 reported units | Baseline 129.19B → Optimized 126.30B

AI Decision Analysis

Measured Comfort Result (EnergyPlus): Degraded

ZONE	CONTROL	CURRENT	RECOMMENDED	LLM ESTIMATED ENERGY SAVINGS	AI CONFIDENCE	LLM PREDICTED COMFORT IMPACT
CORE_MID	Cooling Setpoint	24.00°C	24.50°C	1.00%	90.00%	Likely Improved

ZONE	CONTROL	CURRENT	RECOMMENDED	LLM ESTIMATED ENERGY SAVINGS	AI CONFIDENCE	LLM PREDICTED COMFORT IMPACT
PERIMETER_BOT_ZN_2	Cooling Setpoint	24.00°C	24.50°C	0.50%	90.00%	Possible Worsening



4. Architecture Document

Includes:

- Tool-calling architecture
- Prompt engineering
- Communication workflow
- Simulation handling
- Optimization pipeline

13. GitHub Repository

Repository:

<https://github.com/Prakki-Krishna-Karthik/smart-building-ai-agent>

14. Project Demonstration

Video should demonstrate:

1. Launch Streamlit application
2. Upload/select building model

3. Click **Run Optimization**
 4. AI analyzes simulation outputs
 5. HVAC parameters are updated automatically
 6. New simulation is executed
 7. Dashboard displays quantitative improvements
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15. References

1. **EnergyPlus Documentation**
<https://energyplus.net/documentation>
2. **EnergyPlus Python API**
<https://energyplus.readthedocs.io>
3. **ASHRAE Standard 55 – Thermal Environmental Conditions for Human Occupancy**
<https://www.ashrae.org>
4. **Fanger, P.O. (1970). Thermal Comfort: Analysis and Applications in Environmental Engineering.**
5. **ISO 7730: Ergonomics of the Thermal Environment**

16. Conclusion

The Smart Building AI Agent demonstrates an intelligent closed-loop optimization framework that integrates EnergyPlus with Large Language Models to automate HVAC control. By continuously analyzing simulation outputs, modifying building parameters, and validating thermal comfort using PMV and PPD, the system reduces energy consumption while preserving occupant comfort. This approach eliminates repetitive manual tuning, improves optimization efficiency, and showcases the practical application of AI agents in sustainable smart building management.